smbX.ai — PLATFORM REFERENCE

Single source of truth for decisions, architecture, and standards

Place in repo root. CC reads this alongside CLAUDE.md.

Last updated: March 28, 2026

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§1. THE LINE — What Yulia Can and Cannot Do

smbX does 90% of what an IB does — everything that doesn't require a license. Yulia is the AI deal team (analyst + associate + VP).

Golden Pattern

Analysis → Options → Implications → User Decides.

Every interaction follows this sequence. Yulia presents analysis, lays out options with trade-offs, explains implications of each path, and the user makes the decision.

Yulia ALWAYS Does

- Generate documents (valuations, CIMs, financial models, DD checklists, LOIs, pitch decks)
- Draft communications for the user to send (buyer outreach, counter-offers, DD follow-ups, attorney requests)
- Score and rank offers against objective criteria
- Drive the deal timeline with proactive nudges and status tracking
- Coordinate due diligence (checklist generation, item tracking, red flag identification)
- Model scenarios (base/bull/bear, sensitivity analysis, what-if)
- Present market data with sources cited

Every drafted communication ends with: **"[Review and send when ready — adjust the tone and details to match your style.]"**

Yulia NEVER Does

- **Recommend** — never says "you should" or "I recommend." Says "here are three options" with analysis of each
- **Negotiate on behalf** — never contacts counterparties. Drafts messages for the user to send

- **Advise** — never says "my advice is." Presents analysis and implications
- **Represent** — never acts as the user's agent or fiduciary
- **Guarantee** outcomes, prices, or timelines — gives ranges with methodology, never promises
- **Use "we" language** — Yulia is a tool, not a partner. "I've prepared" not "we've prepared"
- **Price anchor against advisors** — never says "a broker would charge $50K for this." Sells the intelligence, not the price gap

Regulatory Boundary

The platform is NEVER described as an "AI advisor" in any user-facing context due to fiduciary language concerns. It is a "deal intelligence platform." Yulia is an "AI deal team" or "AI analyst."

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§2. PRICING MODEL (Authoritative — March 26, 2026)

Subscription Tiers

Tier	Price	Key Features
Free	$0	Unlimited Yulia conversation + ONE structured deliverable (email required to receive it)
Starter	$49/mo	ValueLens, deal scoring, VRR, SDE/EBITDA analysis, Investment Thesis, Capital Stack, basic exports
Professional	$149/mo	Everything in Starter + CIM, deal room, matching, sourcing, DD/LOI tools, living documents
Enterprise	$999/mo	Everything in Professional + unlimited team members, white-label outputs, API, portfolio dashboard

Rules

- Monthly billing only at launch. No annual pricing until 90 days of churn data.
- 30-day free trial of Professional available to all users.
- No per-deal fees. No success fees. No wallet. **WALLET IS DEAD.**
- No separate broker/advisor tiers. Solo broker = $149 Professional. Broker with team = $999 Enterprise.
- Subscription continues post-close for PMI support (180 days).

- "Contact Sales" does not exist anywhere. All CTAs route to "Talk to Yulia" in chat.
- TEST_MODE=true in env vars → getUserPlan() returns 'enterprise' for all users.
- The free deliverable is per-user, not per-session. Tracked on users table.
- Email capture required for the free deliverable — this is the account creation moment.

Paywall Placement

NOT at a fixed gate (S2/B2). Triggers after the first free deliverable — whenever that happens in the conversation. The upgrade from $49 Starter to $149 Professional triggers when the user needs a Professional-tier feature (CIM, deal room, matching).

Gates Are Readiness-Only

Gates check whether prerequisite work is done. They NEVER check payment status. Gate advancement is always allowed if readiness criteria are met.

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§3. DESIGN SYSTEM — Guber v5.2

Typography

- **Headings:** Sora (weight 800 for page titles, 600 for section heads)
- **Body:** Inter (weight 400 regular, 500 medium, 600 semibold)
- **Monospace:** For financial figures in some table contexts

Colors

Token	Value	Usage
Background	`#FAFAFA`	Page background
Primary text	`#1A1A18`	All body text
Secondary text	`#44403C`	Subheadings, labels
Muted text	`#6E6A63`	Captions, footnotes
Terra cotta	`#D4714E`	**Functional only** — active states, send button, section labels, hero data numbers. NEVER decorative backgrounds.
TerraSoft	`#FFF0EB`	User message bubble background

Token	Value	Usage
Cream	`#FAF8F4`	Alternate section backgrounds
Card bg	`#FFFFFF`	White cards
Card border	`#eeeef0`	Card borders, dividers
App bg	`#f9f9fc`	App shell background

Logo

`smbX.ai` — ONLY the X is terra cotta (`#D4714E`). Everything else (smb, .ai) is primary text color.

Engine brand variant: `smbX.ai Engine` — X and "Engine" are terra, "smb" and ".ai" are primary.

Key Component Rules

- **ChatDock:** ONE shared component used on both public pages and chat. Tools popup, file upload, attachment chips, auto-expand textarea.
- **User message bubbles:** bg `#FFF0EB`, text `#1A1A18`, border 1px solid rgba(212,113,78,0.18), shadow 0 1px 3px rgba(26,26,24,.06). NOT solid terra or dark backgrounds.
- **Desktop hero pattern:** `hidden md:flex min-h-[calc(100vh-80px)] flex-col items-center justify-center px-6` with gap:7vh, max-w-[860px] chat card with shadow-2xl/40px radius/2px solid `#D1D5DB`.
- **iOS Safari:** overflow-hidden must NEVER be applied to position:fixed chat containers. Use `--app-height` from visualViewport.height.
- **Chat morph:** Uses AppPhase state machine with CSS animations — not useNavigate (causes remount, kills fade).

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§4. DELIVERABLE RENDERING GUIDE

Every deliverable has up to three rendering targets:

1. **Canvas** — interactive React view side-by-side with chat (tabbed workspace)
2. **Export** — polished downloadable file (PDF/DOCX/XLSX/PPTX)
3. **Share** — secure read-only view for external recipients

Category A: Narrative Reports → PDF

ValueLens, Business Valuation, Seven-Factor Analysis, VRR, Market Intel, SBA Analysis, DD Risk Summary, Deal Screening Memo, Earnings Summary.

- Server-side PDF via reportlab (Python). Branded cover page, headers/footers, financial tables with IB formatting, charts as embedded PNG.
- **CC skill reference:** `/mnt/skills/public/pdf/SKILL.md`

Category B: Marketing Documents → PDF / PPTX

CIM (10-60 pages, league-adapted), Blind Teaser (1-2 pages), Executive Summary, Pitch Deck (12-20 slides).

- CIM is the flagship. Full TOC, financial exhibits, watermarking for shared copies.
- Pitch Deck uses pptxgenjs.
- **CC skill references:** `/mnt/skills/public/pdf/SKILL.md`, `/mnt/skills/public/pptx/pptxgenjs.md`

Category C: Financial Models → XLSX

Pro Forma, SBA Model, Capital Structure, Working Capital, Cap Table, Dilution Model, Funds Flow, Earnout, Post-Close Cash Flow.

- Real Excel formulas, not hardcoded values. IB color coding: blue inputs, black formulas, green cross-sheet, yellow assumptions. Negatives in parentheses, zeros as "-".
- **CC skill reference:** `/mnt/skills/public/xlsx/SKILL.md`

Category D: Legal/Transactional → DOCX

LOI, Counter Proposal, Term Sheet Analysis, DD Checklist (with actual checkbox fields), Closing Checklist, Data Room Structure.

- Clean letterhead, proper business letter format. Attorneys must be able to edit these.
- **CC skill reference:** `/mnt/skills/public/docx/SKILL.md`

Category E: Operational Plans → PDF / DOCX

Integration Plan (PMI), Day Zero Checklist, Employee Comms, Customer/Vendor Outreach, Transition Plan, Seller Training Schedule.

Category F: Communication Drafts → In-Chat

Buyer outreach, counter-offers, DD follow-ups, attorney requests. Rendered as cards in chat with "Copy" button. Not files.

Brand Template System

All exports share a common `smbxBrand.ts` module:

- Terra cotta as accent only (section dividers, one data highlight — never backgrounds)
- Typography: 24pt title → 18pt section → 14pt subsection → 11pt body → 9pt footnote
- Every doc: branded cover page, headers, footers, confidentiality notice
- Financial tables: right-aligned, $ with commas, negatives in parentheses, alternating rows, source footnotes
- Data always cites source: "Source: U.S. Census Bureau, CBP (2023)"
- White space is generous — spacious = premium
- Disclaimer on final page (not intrusive)

Design Principles

1. Every document has a cover page
2. Every document has proper headers/footers
3. Financial tables use IB formatting
4. Charts are clean — no 3D, no rainbow. Max 3-4 colors. Source citation below.
5. Terra cotta is an accent, not a theme
6. White space is a feature
7. Data always cites its source
8. Every export is self-contained (readable without chat context)
9. Disclaimers present but not intrusive

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§5. INTERACTIVE CANVAS ARCHITECTURE

Tabbed Workspace (Dia Browser Model)

The canvas supports unlimited tabs. Each tab is an independent rendering surface. Tabs persist within a deal session. Users can close, reorder, pin. "+" creates new.

Tab types:

- Model tabs — live interactive (valuation, LBO, SBA, DCF, sensitivity, comparison, cap table, earnout)
- Document tabs — view/annotate (CIM, valuation report, DD checklist)
- Comparison tabs — derived from two or more model tabs (auto-update when sources change)
- Data tabs — sourcing results, market intelligence, pipeline view

Two Input Channels

Every model tab has an `assumptions` state object. Both UI controls (sliders, inputs, toggles) and Yulia (via `update_model` tool) modify the same state → deterministic recalculation → instant re-render.

Critical: calculations are deterministic JavaScript, not AI. DSCR, IRR, MOIC, valuation ranges, amortization — all pure functions. Same inputs always produce same outputs. Instant (<16ms). Auditable. Exportable as real Excel formulas.

AI is used for: generating initial assumptions, interpreting results, suggesting scenarios, drafting communications based on outputs.

Yulia's Canvas Tools

```
update_model      — modify assumptions in any tab (by tabId or "active")
create_model_tab  — open a new interactive model tab
render_to_tab     — push generated content to a tab
read_tab_state    — read assumptions + outputs from any tab ("active" or "all")
```

Cross-Tab Awareness

Yulia's system prompt includes a summary of all open tab states. She can reference any tab. Linked tabs auto-update when source tabs change. Comparison tabs derive from model tabs.

User: "What about this one?" → Yulia reads active tab.
User: "Apply same assumptions to Company B" → Yulia copies Tab 1 to Tab 2.
User: "Which deal should I pursue?" → Yulia reads all tabs and compares.

Interactive Model Types

Valuation Explorer

Sliders for add-backs, multiple range, methodology weights. Live valuation range chart, SDE waterfall, multiple context chart.

LBO / Acquisition Model

Purchase price, EBITDA, growth, exit multiple, debt structure, synergies. Returns dashboard (IRR, MOIC, cash-on-cash), DSCR gauge, pro forma table, sensitivity matrix. This is where "increase EBITDA by 10%" works.

SBA Financing Calculator

Down payment %, loan term, rate, seller note. Go/no-go traffic light (DSCR ⩾ 1.25 = green), monthly payment breakdown, amortization schedule.

Deal Comparison

Reads from other model tabs. Side-by-side cards, radar chart, comparison table with green/red highlighting. The "which deal should I buy" view.

Sensitivity Matrix

Pick two variables, see output metric at every intersection. Color gradient from red (below hurdle) to green (exceeds target).

Cap Table / Dilution (Raise)

Pre/post ownership pie, dilution waterfall, payout waterfall at exit values, MOIC per investor class.

Earnout Scenario

Probability-weighted expected value, fan chart (best/expected/worst), effective total price.

Calculation Engine

Pure JavaScript functions shared client/server:

- `calculateSDE`, `calculateValuationRange`
- `calculateIRR`, `calculateMOIC`, `calculateDSCR`, `buildProForma`
- `calculateSBAEligibility`, `calculateAmortization`
- `calculateDilution`, `calculateWaterfall`
- `buildSensitivityMatrix`

Export from Interactive Models

- PDF: snapshot of current state as branded report. Charts as images. Assumptions in appendix.

- XLSX: full model with real Excel formulas. Blue inputs, black formulas. Sensitivity as separate sheet.
- Version history tracks each export as point-in-time.

Mobile

Full-screen sheet over chat. Horizontal scroll tab strip. Touch-friendly sliders. Back gesture preserves tab state.

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§6. SOURCING ENGINE

Architecture

Research-first: every sourcing journey begins with an Acquisition Intelligence Brief (industry landscape, NAICS mapping, market structure, deal economics, competitive buyer landscape) before any specific targets.

Data Sources

Source	What It Provides	Lag	Cost
Census CBP (2023 endpoint)	Establishment counts, employees, payroll by NAICS × geography	2-3 years	Free
BLS QCEW	Employment, wages, establishment counts by NAICS × county	6 months	Free
Census BDS	Firm age distribution, entry/exit rates	Annual	Free
FRED	Prime rate, unemployment, CPI, regional data	Real-time	Free
SBA 7(a) loan data	Loan counts, amounts by NAICS × state	1 quarter	Free
Google Places	Specific businesses by name, address, rating, reviews	Real-time	~$0.03/request

What's Built (2,374 lines)

marketDataService.ts (Census/FRED/BLS), discoveryService.ts (Google Places), thesisMatchingService.ts, buyerSourcingService.ts, listingIngestionService.ts, searchService.ts,

marketOpportunityService.ts, saleReadinessService.ts, revenueEstimationService.ts, ownershipDetectionService.ts

What's Not Wired Yet

- `scan_market` tool (Yulia can't trigger discovery from conversation)
- Census BDS integration (no firm age data)
- SBA lending dataset (no sba_loan_stats table)
- Website enrichment via Haiku
- Aggregator monitoring (daily BizBuySell scan)
- Market heat scoring service
- Company type classification (see §7)

Multi-Source Display

Yulia should present layered data: "Census says 847 establishments (2023). BLS quarterly data shows 12.4% growth. Google Places identifies 614 with active web presence. SBA funded 31 acquisitions in this market last year." Always show data vintage.

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§7. COMPANY TYPE CLASSIFICATION

The sourcing engine must classify every business it encounters. Census counts ALL establishments — PE-backed roll-ups, public company subsidiaries, franchise units, and chain locations are not acquisition targets for most buyers.

Types to Detect

1. **PE-Backed / Roll-Up Platforms** — already owned by private equity. They're buyers, not targets. Detection: SEC filings, website "portfolio" pages, LinkedIn exec titles ("VP of M&A"), Crunchbase. Hard case: PE add-on that kept original name.
2. **Public Company Subsidiaries** — not acquirable. Map via SEC EDGAR subsidiary exhibits, D&B corporate family trees. A public company may operate 200 local service locations.
3. **Multi-Location Chains & Franchises** — chain locations aren't independent. Franchise units MIGHT be acquirable (franchisee can sell). Detection: same website domain, same phone prefix, Google Places "chain" attribute, FDD data.
4. **Non-Physical-Location Businesses** — consulting, digital, professional services. Invisible to Google Places pipeline. Need alternative data sources: state registrations, SBA loans, professional licensing boards.

Why It Matters

The difference between "847 HVAC companies" (Census) and "~580 independently-owned, acquirable HVAC businesses" (after filtering). Platform credibility depends on this.

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§8. COPY RULES

- **Never say "small business"** — smbX serves ALL deal sizes, $300K to mega-cap
- **Advisors/brokers are CUSTOMERS, not competitors** — never position against traditional advisory
- **Never use "old way vs new way" comparisons** — never anchor pricing against advisory fees
- **No deliverable names on journey pages** — lead with outcomes, not product names
- **No "we" language** — Yulia is a tool, not a team member
- **IB comparison framing** — approved for How It Works and Pricing pages only, with mandatory callout clarifying it is not anti-advisor positioning
- **Copy voice** — sounds like a veteran M&A advisor who has closed deals around the world. Human, sympathetic, authoritative, specific. Never AI-generated marketing copy.
- **ValueLens** is the authoritative name (not Bizestimate, not "business valuation tool")
- **"Talk to Yulia"** replaces "Contact Sales" everywhere

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§9. FOUR JOURNEYS — GATE MAP

Sell Journey (S0-S5)

6-month to 2-year arc: understand → optimize → prepare → exit.
Exit types: full sale, partner buyout, capital raise, ESOP, majority share sale, partial stock/asset.

Gate	Name	Key Deliverables
S0	Getting Started	Business profile, league classification
S1	Financial Deep-Dive	SDE/EBITDA analysis, add-back schedule, ValueLens
S2	Valuation & Positioning	Valuation report, seven-factor, market snapshot
S3	Deal Materials	CIM, blind teaser, buyer list
S4	Buyer Matching	Outreach strategy, meeting prep, LOI analysis
S5	Closing & Transition	DD coordination, working capital, funds flow, transition plan

Buy Journey (B0-B5)

Extends 180 days post-close with PMI. Buyer value = speed to conviction.

Gate	Name	Key Deliverables
B0	Thesis & Readiness	Investment thesis, target criteria, capital stack
B1	Deal Screening	Market scan, deal scoring, screening memos
B2	Valuation & Modeling	Financial model, SBA analysis, LOI draft
B3	Due Diligence	DD checklist, risk summary, QoE
B4	Deal Structure	Sources & uses, earnout model, working capital
B5	Closing & Integration	Funds flow, closing checklist → auto-transition to PMI

Raise Journey (R0-R5)

Gate	Name
R0	Readiness Assessment
R1	Financial Package
R2	Investor Materials (pitch deck, data room)
R3	Investor Outreach

Gate	Name
R4	Term Negotiation
R5	Closing the Raise

PMI Journey (PMI0-PMI3)

Gate	Name
PMI0	Day Zero (first 48 hours)
PMI1	Stabilization (30 days)
PMI2	Assessment (60-90 days)
PMI3	Optimization (90-180 days)

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§10. WHAT'S BUILT vs WHAT'S NOT (Post-Cleanup, March 28, 2026)

Built and Working

- Chat loop: anonymous + authenticated, SSE streaming, morph from landing to chat
- 12 agentic tools in tools.ts
- 28 generators with model routing (deterministic/haiku/sonnet/opus)
- 45 normalizer switch cases + 10 aliases + category generators
- Gate system: 22 gates, readiness checks, gate prompts (1,171 lines)
- Knowledge injection: industry profiles, SBA rules, methodology, market context
- Seven-factor scoring (123 lines)
- All 7 journey pages as content components (6,434 lines total)
- AppShell (1,516 lines) with unified routing, deal-grouped sidebar, canvas
- Export pipeline: PDF/DOCX/XLSX (functional — being upgraded to branded templates)
- Data room with folders, share links, NDA, access logs
- Collaboration RBAC: owner/attorney/CPA/broker/lender roles
- Sourcing infrastructure: Google Places, Census CBP, thesis matching, listing pipeline
- Auth: register, login, JWT, password reset, anonymous→auth migration

- pg-boss workers: daily thesis scan, quarterly refresh

Not Built Yet

- subscriptionService.ts (subscription access checks — Session A prompt ready)
- Interactive canvas models (architecture designed, components being built)
- Living documents (stale detection, regeneration, version tracking)
- Auditor mode (prompt injection when processing uploaded docs)
- Market heat scoring service
- scan_market tool for Yulia
- Website enrichment service
- Aggregator monitoring
- Company type classification
- Professional branded export templates (CC working on this NOW for ValueLens)
- draft_communication tool

Recently Cleaned Up (March 28, 2026)

- All wallet code removed
- Bizestimate renamed to ValueLens everywhere
- Pricing pages updated to $49/$149/$999
- Advisor-specific tiers removed
- Orphan files deleted
- Census CBP endpoint updated to 2023
- CLAUDE.md rewritten